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Assignment-2

Section-A

1. The Entity-Relationship (ER) Diagram visually represents how entities relate in database, offering a clear blueprint of the data's organizational structure.

2. In the relational model data is organized into tables, with relationships between entities defined through foreign key constraints.

3. The object-oriented model treats data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions.

4. Integrity Constraints ensures the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Section-B
(Case-Study based)

Q-2 In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professional and patient records.

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Ans-2 Data abstraction in a healthcare database helps hide complex database structure and presents only the necessary info to users. It simplifies how healthcare professionals interact with patient records:

There are three levels of data abstraction:

1. Physical level:

This level describes how data is actually stored in the database. Healthcare professionals do not need to understand storage details such as files, indexes, or storage locations.

2. Logical level:

This level defines what data is stored and the relationships between the data.

for example: patient records may include patient ID, name, medical history, diagnosis, prescriptions, and doctor details.

3. View level:

This level allows different users to see only relevant data.

for example:
- Doctor can view diagnosis, medical history
- Nurses can see treatment schedules and patient
- Administrative staff can access billing and appointmen

by data abstraction, healthcare system become easier to use, more secure, and more efficient for managing patient info.

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Ques-3 Explore the potential benefits and challenges associated with implementing data abstraction in this content?

Benefits

1. Improved data Security
Sensitive data patient can be protected by restricting access based on user roles.

2. Simplified User Interaction
Healthcare professionals interact with simple interfaces without worrying about complex database structure

3. Data Consistency
Abstraction ensures that the same data structure is used across the system, reducing errors and duplication.

4. Better System Maintenance
Changes in database structure can be made without affecting users or applications.

5. Efficient data Management
Patient data records can be managed, update, and retrieved quickly.

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Challenges

1. System Complexity
Designing abstraction layers requires careful planning and skilled developers.

2. Performance Overhead
Multiple abstraction layers may sometimes slow down data access.

3. Implementation Cost
Developing secure and well-structured healthcare databases may require significant investment

4. Training Requirements
Healthcare staff may need training to use the system effectively.

SECTION-C
(Simulation Based)

Ques-3 Design a database for a university, incorporating entities like students, courses and professors. Using the ER diagram, illustrate and explain the relationships between these entities. Include attributes for each entity to capture relevant information such as student names, course details, and professor information.

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A university database system manages information about students, courses, and professors.

Entities and attributes

1. Student
- Student_ID (PK)
- Name
- Email
- Department
- Year

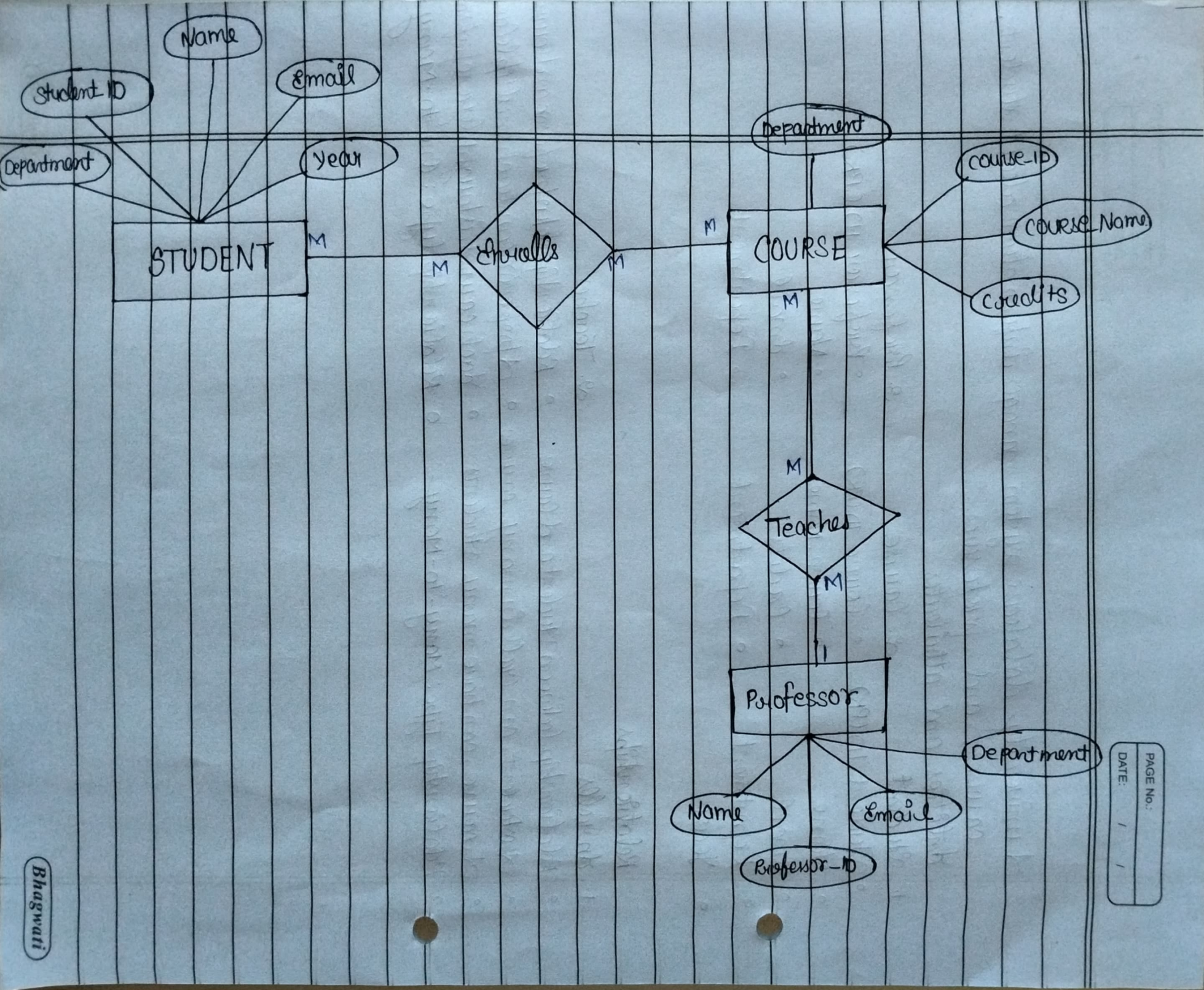
2. Course
- Course_ID (PK)
- Course_Name
- Credits
- Department

3. Professor
- Professor_ID (PK)
- Name
- Email
- Department

Relationships

1. Enrolls
- Relationship between Student and Course
- A student can enroll in many courses
- A course can have many students
- Relationship Type: Many-to-Many

2. Teaches
- Relationship b/w professor and course
- A professor can teach multiple cou
- Each courses is taught by one profess
- Relationship type: One-to-Many



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